BaiZe: A 2.2B Local Multimodal Subagent for EDA Scripting

and a Test of the Observation-Fidelity Account

***Abstract*—ZhuLong established that, for execution-grounded EDA-scripting agents, accuracy is bounded by *observation fidelity* ε rather than by model scale: with the harness fixed, six backbones span only 23.7 pp, whereas the observation axis spans 57.2 pp. This paper asks the two questions that conclusion leaves open. Goal 1 (cost). Can a ≈2.2B hybrid model, served on-premises, reach the accuracy of the weakest measured backbone (53.2%) on the same 449-task benchmark while cutting per-task serving cost? Goal 2 (observation). Does adding a visual channel—layout, schematic and waveform views, which make up 70.2% of the benchmark—shrink the grounding gap, and at what visual-token budget? Our answer takes the form of three falsifiable claims, each bound to a pre-registered experiment: A a 2.2B backbone matches the 53.2% floor at materially lower cost; B the 196-token view used by our encoder may *widen* the gap on detail-dense views, with a measurable token threshold; C the read-back ladder (N′ →B→F, 7.1 pp on ZhuLong) can be re-measured on the small model, so that a *larger* ladder span supports the ε-bound and a smaller one would demand the opposite reading. The system is built from scratch: stage (i) language pre-training (2.94T tokens, 700k steps at sequence $16{,}384$ on $16\times$H100; iteration $30{,}660$ at loss 2.151, ≈598K tok/s) and stage (iii) encoder pre-training (126.8M parameters, $224/16$ patches, 196 visual tokens, 19.76% / 20.27% ImageNet linear probe at 55.3M / 176M samples) are *complete*; stages (ii), (iv) and (v) are given as protocols with placeholder tables, expected directions and reverse contingencies. We report each claim's verification status explicitly instead of presenting an unmeasured direction as a result.**

***Index Terms*—small language model, agentic LLM, hybrid state-space architecture, observation fidelity, EDA scripting agent, multimodal grounding, serving economics**

I. Introduction

ZhuLong [1] closes the loop on EDA scripting: four MCP tools (semantic API search, keyword search, documentation lookup, sandboxed execution) let a large language model generate PyAether, SKILL and Tcl code, run it, read the state back, and refine. On a 449-task benchmark it reports 76.9% pooled Pass@1, against 19.7% for the same model without any harness. Its central finding is negative for scaling: with the harness held fixed, six backbones span 23.7 pp (76.9% to 53.2%), while the observation axis—what `run_code` returns—spans 57.2 pp. Accuracy, on this account, is a property of *observation fidelity* ε, not of parameter count.

That conclusion leaves two engineering questions unanswered, and they are the ones a deployment team actually faces. **First, cost.** ZhuLong's full configuration spends ≈710k input tokens, 10.6 `run_code` calls and 464 s of net-active time per task; removing retrieval pushes the time to 981 s ($2.1\times$). At fleet scale these are the dominant terms, and proprietary layouts, schematics and review judgements cannot leave the premises anyway, so the alternative to a local small model is not a hosted large model but no model at all. For an agentic session the binding resource is not prompt throughput but the state that grows with accumulated context; in a hybrid backbone only the four attention layers carry that state, while the 52 Mamba-2 layers hold a constant-size recurrent one. **Second, observation.** Layout and schematic views account for 70.2% of the benchmark's tasks, yet ZhuLong is a text-only agent: it can manipulate these artifacts only through scripts, never look at them. Waveforms are the same story—period, edge count, glitch and threshold crossing are quantities that text represents poorly.

a) This paper.: We build BaiZe, an end-to-end system trained from scratch in five stages—(i) language pre-training, (ii) agentic post-training, (iii) vision-encoder pre-training, (iv) multimodal alignment, (v) semiconductor-domain adaptation—and use it to answer both questions. Stages (i) and (iii) are complete; stages (ii), (iv) and (v) are specified as protocols whose measurements are placeholders. We state this openly because the contribution we claim is not a finished leaderboard entry but a set of *testable* claims with pre-registered failure conditions.

b) Claims.: **A (efficiency).** Under ZhuLong's harness, the 2.2B hybrid reaches pooled Pass@1 no lower than the weakest measured backbone (53.2%), at materially lower per-task serving cost. **B (observation).** The visual channel shrinks the grounding gap on detail-dense EDA views only above a token budget; the 196-token view may widen it. **C (theory).** The read-back ladder N′ →B→F is re-measurable on the small model, and its span relative to ZhuLong's 7.1 pp decides whether ε binds harder on small models (larger span, supporting the ε-bound) or is partly replaced by weight priors (smaller span). Claim C is the only place where a continuation of ZhuLong can *test* ZhuLong rather than merely apply it; we therefore reserve RQ4 for it.

c) Contributions.: (i) A reproducible from-scratch 2.2B hybrid backbone (2.94T tokens, 700k steps, sequence $16{,}384$) with a published schedule and cost accounting. (ii) A completion protocol for the agentic stack—domain annealing, SFT, RLVR and preference-based domain adaptation—specified so that only the backbone varies against ZhuLong's byte-identical harness (Table IV). (iii) A pre-registered multimodal study of visual-token budgets on layout, schematic and waveform

views, including a waveform-reading task drawn from ZhuLong's WaveView failures (Section VI). (iv) **A test of the ε-bound itself** (RQ4): a re-measurement of the read-back ladder on a 2.2B model, with both outcomes interpretable in advance.

d) Positioning.: "ZhuLong answers what bounds the capability of an EDA agent; BaiZe asks whether, at fixed ε, a 2.2B on-premises model plus a visual observation channel can reach a deployable ε' at acceptable cost—and whether ε' contracts or expands relative to ε." We do not claim that a 2B model "matches" large models; the comparison we pre-register is against the *weakest measured backbone*, and a miss converts Claim A into a cost–accuracy Pareto report rather than a failure to publish.

II. Background and Related Work

A. Small Language Models and Agentic Reasoning

Parameter-efficient SLMs ($< 2\,\mathrm{B}$) have recently narrowed the gap with larger models on mathematical and commonsense reasoning. MiniCPM [2] introduced scalable training strategies and domain-weighted data mixing for small models; Qwen [3] and Gemma [4] demonstrate that strong code- and math-enriched corpora can lift small models substantially. Most of these efforts, however, treat the final trained model as the deliverable and report comparatively little about the *search* that produced it. Complementarily, agentic instruction-tuning datasets (e.g., tool-use and multi-step reasoning corpora) [5] have been shown to elicit planning and tool-invocation behaviour, yet are rarely paired with a from-scratch pre-training recipe for the 1B regime.

B. Hybrid and State-Space Architectures

State-space models (SSMs) such as Mamba [6] and Mamba-2 [7] replace full attention with a recurrent scan update, offering linear-time decoding that suits inference-bound deployments. Hybrid designs interleave a small number of attention layers among SSM/MLP layers—notably Nemotron-H [8] and Hymba [9]—to recover long-range context while retaining fast decode. Released Megatron-Core variants of these families are scarce below 2B, leaving their small-scale trade-offs under-documented; our controlled 2B comparison (Section IV-A) addresses precisely this gap.

C. Multimodal Vision-Language Models

Vision-language models connect a vision encoder to a language model through a lightweight projector, enabling end-to-end visual understanding and instruction following. LLaVA [10] established visual instruction tuning—aligning a vision encoder to a frozen LLM via a linear projection followed by instruction fine-tuning—as a scalable recipe for open multimodal assistants. LLaVA-OneVision [11] consolidates this line into a single open model spanning single-image, multi-image, and video tasks through a staged curriculum: stage-1 projector pre-training, stage-1.5 mid-training, and stage-2 instruction tuning. BaiZe follows this curriculum in stage (iv) (Section VI-B), substituting a from-scratch vision encoder (stage iii) for the frozen off-the-shelf encoder.

D. LLMs for Electronic Design Automation

A growing line of work applies LLM agents to chip design tasks such as register-transfer-level generation, verification, and bug triage [12], [13]. Concurrent work has begun to benchmark EDA agents and to drive EDA tools directly; we do not re-benchmark, and deliberately reuse one published harness rather than propose another, because our claims concern *what varies* inside a fixed harness.

E. The Relation to ZhuLong

BaiZe is a continuation, not a competitor: both papers read accuracy through the same observable—assertion-verified script execution—and both hold the scaffold fixed while varying one axis. ZhuLong varies the observation and prior axes and concludes that fidelity bounds accuracy; BaiZe varies the *scale* axis and, in stage (iv), the *channel* axis—which is what makes its result a test rather than an application. Two of ZhuLong's findings constrain what we may claim. First, component contributions are toolchain-specific—retrieval is worth 15.1 pp on SKILL but 7.7 pp on Tcl—so PyAether alone cannot support a generality claim, and we treat SKILL as the stress slice. Second, the read-back ladder inverts at the middle rung on Tcl because short scripts are near-single-shot; we therefore pre-register the ladder itself rather than an expected ordering. Because layout and schematic views make up 70.2% of the benchmark while ZhuLong's agent is text-only, the visual channel is the one axis ZhuLong could not vary and BaiZe adds.

III. System Architecture

BaiZe is assembled in five stages: (i) from-scratch pre-training of a hybrid language backbone, (ii) agentic post-training (annealing, SFT, RLVR), (iii) from-scratch pre-training of a vision encoder, (iv) multimodal alignment, and (v) semiconductor-domain adaptation. This section fixes what is shared—the two backbones, the serving-state argument, the subagent protocol, the visual-token interface, and the harness contract—before each stage is specified in its own section.

A. Language Backbone

The backbone is a sequential hybrid [7], [8]: 56 layers interleaving Mamba-2 state-space layers, four grouped-query-attention (GQA) layers and MLP layers, reusing Megatron-Core's 56-layer Mamba-hybrid pattern verbatim while halving only the hidden and FFN widths. It is not a Nemotron-H model: an order of magnitude smaller, no mixture-of-experts, pre-trained from scratch on an independent corpus under a fixed GPU-hour budget. Each Mamba-2 layer uses 8 state-space groups; the attention layers use 16 query heads and 4 query groups with untied embeddings (Table I). Only the four attention layers carry rotary position embeddings, so long-context extrapolation depends on $4/56 = 7.1\%$ of layers (Section IV-H).

TABLE I
LANGUAGE-BACKBONE CONFIGURATION (MAMBA2-HYBRID, ≈2.22B).

Hyper-parameter	Value
Transformer layers	56
Architecture	hybrid (Mamba-2 + GQA + MLP)
Attention layers	4 (interleaved)
Hidden size	2048
FFN hidden size	8192
Attention heads (Q) / groups	16 / 4
Mamba-2 state-space groups	8
Vocabulary size	129,281
Sequence length	16,384 (production)
Normalisation / activation	RMSNorm / SiLU
Parameters	≈2.22B

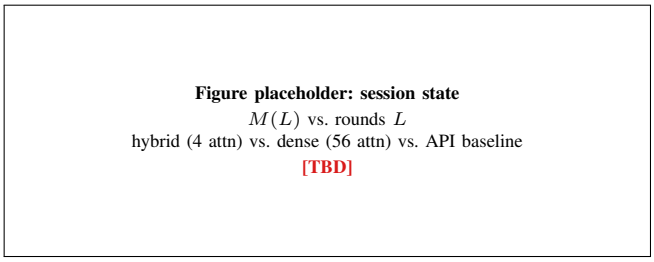


Fig. 1. Per-session serving memory against accumulated context. Predicted shape from Eq. 1: the hybrid grows $14\times$ slower in the attention term and adds a constant SSM term. Measurement is pending.

B. Why a Hybrid: Constant-Size Session State

An agentic session is not a prompt: each `run_code` round appends code, stdout, trace and error text, and the conversation that results is re-read on every subsequent step. For a dense Transformer the cached key/value state grows linearly in the session length, so per-step memory is $O(L)$ and total cost is quadratic in the number of rounds. In BaiZe only the four GQA layers contribute a growing cache; the 52 Mamba-2 layers carry a recurrent state of fixed size. Writing $L_a{=}4$ attention layers with h KV heads, head dimension d, context L and BF16 storage, the session state is

$$M(L) \;=\; \underbrace{2\,L_a\,h\,d\,L}_{\text{attention cache}} \;+\; \underbrace{C_{\text{ssm}}}_{\text{constant}}, \tag{1}$$

where C_{ssm} is independent of L; the corresponding dense model replaces L_a by 56 and therefore grows $14\times$ faster. This is a statement about scaling, not about absolute size, and it is measured, not assumed: Figure 1 is the placeholder for the measurement we will report.

C. Subagent–Master Protocol

BaiZe is not meant to replace the orchestrator; it is meant to be one expert per tool, reachable from a stronger planner or from a human flow (Figure 2). The boundary is deliberately asymmetric: the subagent exchanges *natural language* with its caller and *EDA scripts* with the tool, so a master never has to understand SKILL or Tcl, and the subagent never has to trust the master's code. Table II fixes the message schema; every field is machine-checkable, which is what lets us swap the backbone without touching the harness.

TABLE II
SUBAGENT–MASTER MESSAGE SCHEMA (T3). →: CALLER TO SUBAGENT; ←: SUBAGENT TO CALLER.

Field	Type	Constraint	Dir.
`role`	enum	`layout` / `schematic` / `waveform`	→
`intent`	text	NL request, ≤512 tok	→
`artifacts`	image[]	rendered view(s), 224–448px	→
`budget`	int	max `run_code` calls	→
`script`	code	PyAether / SKILL / Tcl	←
`tool_calls`	record[]	tool id + args (audit)	←
`assertion`	record	per-assertion pass/fail	←
`handoff`	text	NL summary + open issues	←

Every field is validated at the tool boundary, so the caller cannot inject execution and the subagent cannot claim success without an assertion record.

TABLE III
VISUAL-TOKEN TIERS EXPOSED BY THE STAGE-(IV) INTERFACE (T4).

Tier	Input	Tokens	px/token	Rel. cost
T196 (default)	224	196	16×16	1.0×
T576	336	576	16×16	2.9×
T1024	448	1024	16×16	5.2×

Rel. cost counts visual tokens only; end-to-end serving cost per task is measured separately (Table XIX). Which tier is "enough" for layout, schematic and waveform views is the object of RQ5–RQ6.

D. Visual-Token Interface

The encoder of stage (iii) renders an image as $14{\times}14{=}196$ tokens at 224px/16-px patches; two coarser-finer tiers are pre-registered so that Claim B is measured rather than argued (Table III). The tier is a single scalar in the projector, so accuracy and cost move along one axis and the ablation cannot confound architecture with resolution.

E. Harness Contract: Byte-Identical

Because ZhuLong attributes its gains to the interface rather than to the policy, varying the backbone is only informative if everything else is frozen. Table IV is the checklist we commit to; any deviation invalidates the comparison and is reported as such.

IV. LLM PRETRAINING

Stage (i) pre-trains the hybrid language backbone (Section III-A) from scratch under a fixed budget.

A. Architecture Selection

We adopt the hybrid backbone for three reasons: (i) state-space layers decode in constant time per token while the four attention layers retain long-range context; (ii) a matched from-scratch comparison at the ≈2B scale against a dense transformer (MiniCPM5-2B [14]) favours the hybrid, which reaches lower loss with 11.6% fewer parameters (≈2.22B vs. 2.51B) at a ∼8% training-throughput cost (Table VIII); and (iii) Megatron-Core ships a Mamba-hybrid provider whose 56-layer interleaving pattern we reuse verbatim, making the backbone directly executable without custom kernels.

TABLE IV
HARNESS CONTRACT FOR THE BAIZE–ZHULONG COMPARISON (T5).

Component	ZhuLong	BaiZe
Agent scaffold	Cline-CLI, unmodified	identical
MCP tools	4 (search, detail, doc, run)	identical
Retrieval stack	API + keyword + docs	identical
Sandbox / effector	vendor runtime	identical
Anti-cheat hook	enabled	identical
Assertion criterion	per-task assertions	identical
Timeout	2500 s / task	identical
Runs	5, reshuffled, fresh ctx	identical
Contamination	n-gram blacklist	identical
Task schema / slices	158/157/134	identical

Only the backbone (and, in stage iv, the visual channel) varies. Deviations found during the run are listed in the appendix rather than silently corrected.

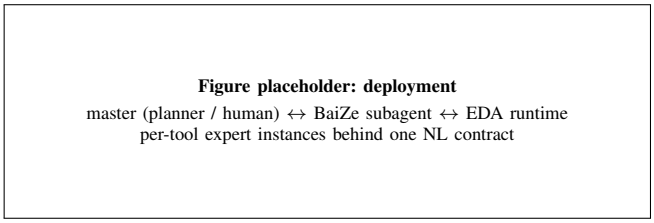


Fig. 2. Deployment shape: one subagent per EDA tool, addressed in natural language, speaking EDA script to the tool.

B. Training Framework and Learning-Rate Schedule

We train with the NVIDIA NeMo [15] entry point driving Megatron-Core's [16] Mamba-hybrid provider at `tensor_model_parallel_size` $= 1$, with fused kernels and selective activation recomputation enabled; all hyper-parameter searches run at a reduced iteration count, with the short-horizon ranking transferred to full scale.

BaiZe trains under a warmup–stable–decay (WSD) schedule with a stable learning rate of 1×10^{-3} and 2,000 warmup steps (0.29% of the 700k total; the search of Section IV-F found 250 optimal at 5k steps, but the proportion narrows at full scale). The main training sequence length is 16,384: this is the throughput-optimal operating point (613,992 tok/s at CP=1/DP=16; measured) and it keeps the extrapolation ratio to the 1M product target at 64× rather than the 256× a 4K length would imply. The decay phase has two segments: segment A anneals at 16K on instruction and curated web data (60–70% of the decay budget), and segment B extends the length on a 32K→64K→128K ladder (30–40%). Decay follows $lr(p) = lr_{\text{peak}}(lr_{\min}/lr_{\text{peak}})^p$ with $lr_{\min}$=1×10^{-4}.

C. Search Protocol

Because a full grid is infeasible, the recipe was selected with a staged, single-variable-at-a-time protocol at a 5000-step horizon: S1 baseline; S2 learning-rate sweep (2×10^{-4}–1×10^{-3}, seven points); S3 schedule axes (WSD vs. cosine, decay ratio, warmup, min_lr); S4 annealing data mix; S5 long run plus three seeds. The total search budget is ≈82 GPU-hours on eight H100s, and the reported losses are proxy measurements for ranking, not production estimates.

TABLE V
MEASURED DOCUMENT-LENGTH DISTRIBUTION OF EN_V1_4 (8 SHARDS, 8.59B TOKENS, 9.66M DOCUMENTS).

Threshold	Docs	Token share	Status vs. target
≥32K	4,257	**2.52%**	target ≥20%: **not met**
≥128K	—	**0.03%**	target ≥5%: **not met**
≥1M	—	not estimated	not required (extrapolated)

D. Data

The stable phase draws on a seven-source mixture—`en_v1_4` (0.65), `ufw_l1_en_hq` (0.05), `ufw_zh` (0.04), `math_l1` (0.05), `math_l3` (0.02), `code_l2` (0.15), and `code_l3` (0.04)—which yields a 74:19:7 web/code/math split (0.74 `Ultra-FineWeb`, 0.19 `UltraData-Code`, 0.07 `UltraData-Math`); a 400-trial Bayesian search at 18.36M proxy scale confirmed this region as optimal within 0.6 pp. Sources are pre-blended into large shards so each rank opens $\mathcal{O}(10)$ index files instead of $\mathcal{O}(600)$, which bounds the data-loader footprint and is a precondition for the throughput above. Contamination is controlled by a 193,295-entry 13-gram fingerprint blacklist built from all 536 benchmark tasks; every shard is scanned for ≥0.80 n-gram overlap before release. The corpus deliberately contains *no* EDA artifacts: no PyAether, SKILL or Tcl API reference, no tool documentation and no vendor script enters pre-training. This is a design choice rather than an omission, and it is what makes the comparison with ZhuLong informative: ZhuLong supplies API knowledge through retrieval at inference time, whereas any such knowledge BaiZe holds must be installed in stage (ii) and carried in weights—the assumption that RQ2 (Section V-C) tests directly.

E. Long-Context Data Gate

Segment B learns whatever length structure the corpus contains, so we measured the document-length distribution before budgeting it: only **2.52**% of tokens sit in documents of ≥32K and 0.03% reach ≥128K, with a mean document of 889 tokens (Table V). Training a 128K rung on this distribution would teach document concatenation rather than long-context retrieval, so we impose three entry conditions: supplement long documents until the ≥32K share reaches 20% and the ≥128K share reaches 5% (until then the ladder stops at 64K); reset attention and Mamba-2 state at every EOD; and de-duplicate long documents first.

F. Results and Production Run

The search selects a stable learning rate of 1×10^{-3} under WSD with a 86:10:4 annealing mixture, reaching a validation loss of **2.2054** at 20000 steps and 2.6739 ± 0.0469 across three seeds at 5000 steps (Table IX). Each ablation confirms a single-variable choice: WSD beats cosine (2.7679 vs. 2.8155), a 10% decay ratio beats 5%, and 250 warmup steps beat 2000.

TABLE VI
PRODUCTION BUDGET. WALL-CLOCK IS ESTIMATED AT MEASURED THROUGHPUT.

Phase	Tokens	Steps	Wall-clock	Status
Warmup + stable	2.94T	700k	≈56 d	30.7k done
Decay A (16K, 60–70%)	58.7B	14k	≈1.2 d	queued
Decay B (LC, 30–40%)	25.2B	6k	≈2.3 d	gated
Total	**3.02T**	**720k**	≈59.5 d	—

The recipe is scaled to 2.94T tokens in the warmup-plus-stable phase at TP=1/CP=1/DP=16 over two 8×H100 nodes, micro-batch 1 at sequence 16,384 (GBS=256, 4,194,304 tokens/step), FP8 tensorwise, computing-graph replay disabled, AdamW. Replay is disabled deliberately: a replayed graph over the data-dependent Mamba-2 state hit a throughput plateau, and this is a precondition for the rate below. At the time of writing (31 October 2026) the run has passed iteration 30,660 of 700,000 at a sustained ≈598K tok/s with training loss 2.151, gradient norms stable at 0.09–0.10, and no NaN; the remaining schedule is projected at the measured rate (Table VI). Precision was chosen per configuration rather than globally: at sequence 4096 the FP8 tile (M=8192) sits below the crossover point and runs at 0.91× BF16, whereas at M=16,384 FP8 pays, and the 613,992 tok/s peak is an FP8 measurement.

G. Long-Context Training in the Decay Tail

Long-context training is segment B of the decay phase: the learning rate is already low, so the length-distribution shift perturbs near-converged weights; and neither SFT nor RL establishes ground retrieval ability at this token scale. The decay phase is budgeted at ∼2.9% of the main training volume (83.9B tokens over 20,000 steps), of which long-context data takes 30–40% (25–34B), split 40–50%/30%/20–30% over the three rungs. The 10–15% range (294–441B) remains a design ceiling should the data gate (Section IV-E) require more long-context exposure. Synthetic needle data takes 20–40% of the long-context tokens and is diversified over single-needle NIAH, multi-needle MK-NIAH (all-hit scoring), multi-hop tracing, variable-length needles, and paraphrase-plus-noise; the remainder is natural long documents, since a purely synthetic segment teaches answering about inserted needles rather than reading documents.

The ladder runs 32K→64K→128K with CP $\in \{2, 4, 8\}$, holding seq/CP = 16,384 per GPU and 4.19M tokens/step at every rung (Table VII). The learning rate continues the decay and is never re-warmed; θ=10^6 is fixed at initialisation, so 128K is reached by training and 1M is only an 8× extrapolation; no position interpolation, NTK rescaling, or YaRN is enabled. The reference rung-cycle costs an estimated ∼880 GPU-hours on sixteen GPUs (Table VI), carried by context parallelism rather than extra nodes. When the WSD schedule ends we take one snapshot and measure base capabilities—common-sense reasoning, needle retrieval (passkey and RULER, target ≥95% at 1M), perplexity extrapolation (≤ 1.10× at 1M), and basic format following—reported separately from post-training. Retrieval uses a depth-by-position grid with all-hit scoring, because perplexity alone is not accepted as evidence of long-context ability; complex reasoning and agentic ability are recorded as a baseline only.

TABLE VII
LONG-CONTEXT LADDER. TOKENS PER GPU ARE HELD AT 16,384; 32K THROUGHPUT IS MEASURED, 64K/128K EXTRAPOLATED (ESTIMATED).

Rung	Seq.	CP / DP	GBS	Tokens	Throughput
1	32K	2 / 8	128	≈42B	351,526
2	64K	4 / 4	64	≈32B	≈180K (est.)
3	128K	8 / 2	32	≈32B	≈90K (est.)

H. Long-Context Capability and Inference Cost

With adjusted base frequency (θ=10^6) on the four attention layers, perplexity evolves 83.2 → 51.5 → 53.3 → 53.0 → 55.8 → 55.4 from 4K to 1M, i.e. flat beyond 64K; the stability follows because only 4 of 56 layers carry RoPE, so the extrapolation burden falls on 7.1% of the network. Under the SGLang production stack, the hybrid achieves 39,377 tok/s prefill at 128K and serves 1M in 172s at 34.7 GB, whereas the dense transformer cannot serve beyond 512K on a single 80 GB H100 (KV-pool cap ≈455K tokens); the hybrid's advantage is architectural, since the Mamba-2 state grows in $O(1)$ for 52 of 56 layers.

TABLE VIII
ARCHITECTURE-LEVEL COMPARISON MOTIVATING THE HYBRID CHOICE (DENSE = MINICPM5-2B, HYBRID = BAIZE; MEASURED FROM SCRATCH UNDER MATCHED DATA, SEED, AND GPU FOOTPRINT, SAME NODE, SAME CONFIGURATION). TRAINING THROUGHPUT WAS RE-MEASURED ON A SINGLE 8×H100 NODE AFTER AN INITIAL CROSS-NODE MEASUREMENT WAS FOUND TO OVERSTATE THE GAP. THE DECODE GAP IS MEASURED UNDER THE MEGATRON-CORE DIRECT-DRIVE HARNESS WITHOUT CUDA GRAPHS, WHERE THE DENSE BASELINE IS CPU-LAUNCH-BOUND, AND THEREFORE NARROWS UNDER PRODUCTION INFERENCE STACKS.

Metric	Dense	Hybrid	Edge
Parameters	2.51B	**2.22B**	hybrid (−11.6%)
Final loss (1k steps)	4.81	**4.68**	hybrid (−0.12)
Train loss (5k steps)	3.36	**3.06**	hybrid (−0.30)
Training tok/s	**90K**	83K	dense (+8%)
Prefill tok/s	21.4K	**23.7K**	hybrid (+10.5%)
Decode tok/s	2.2	**22.8**	hybrid (+10.3×)

TABLE IX
HYPER-PARAMETER AND DATA-MIX SEARCH OUTCOMES AT A MATCHED 5000-STEP (S5: 20000-STEP) HORIZON.

ID	Change vs. best	Val. loss	Outcome
S1-01	baseline (lr=3×10^{-4}, WSD)	2.849	baseline
S2-07	stable LR = 1 × 10^{-3}	**2.7627**	best LR
S3-02	schedule family: cosine	2.8155	WSD retains
S3-03	decay ratio 5%	2.7900	10% retains
S3-04	warmup 2000 steps	2.7957	250 retains
S4-03	anneal L3+code+math 86:10:4	**2.6298**	best mix
S5-01	winner, 20000 steps	**2.2054**	**winner**

V. Post-Training for EDA Agentic Capability

Stage (ii) is where the domain prior enters: the pre-training corpus of Section IV-D is general web, code and mathematics text and contains no EDA artifacts, so every EDA fact the model holds must be installed here. This is also what makes the small model interesting: ZhuLong supplies API knowledge through retrieval, whereas BaiZe can, in principle, carry some of it in weights—a difference RQ2 measures directly. The stage has three steps—domain annealing, supervised fine-tuning and RLVR—and terminates in the preference-based adaptation of Section V-G.

A. Common Protocol

All variants are scored on **EDA-Eval-PyAether** [1] and, where stated, on the full 449-task suite of that benchmark (158 PyAether, 157 SKILL, 134 Tcl, the latter pooling WaveView 77 and Innovus 57). We follow the benchmark protocol exactly—five runs, reshuffled order, fresh context, mean Pass@1 with run-to-run s.d., 2500 s per task—and hold the harness byte-identical (Table IV). A second control is contamination: the 13-gram blacklist of Section IV-D covers 193,295 entries drawn from 536 tasks, and any task whose overlap exceeds threshold is reported rather than dropped silently.

B. RQ1: Does a 2.2B Backbone Clear the Floor? (Claim A)

Table X is the pre-registered comparison: the six backbones ZhuLong measured, plus one row for BaiZe. The question is narrow and falsifiable—*does the 2.2B model reach* 53.2% *pooled*—because ZhuLong's own data show that a 23.7 pp spread separates the strongest from the weakest large backbone, and claiming more than the weakest would be claiming the model beats models $30\times$ its size.

Two outcomes are both publishable and both are pre-registered: clearing 53.2% supports Claim A, while falling short yields the Pareto frontier of Section VII with the accuracy gap quantified per slice. We add one diagnostic that ZhuLong did not need: per-slice deltas, because a small model may hold up on PyAether (where execution can substitute for retrieval) and collapse on SKILL (where it cannot).

C. RQ2: Does a Small Model Need Retrieval More?

ZhuLong reports that removing retrieval costs 10.7 pp pooled, but that the loss is toolchain-specific (15.1 pp on SKILL, 9.1 pp on PyAether, 7.7 pp on Tcl) because PyAether traces can partly rebuild API knowledge by introspecting the sandbox. A 2.2B model has fewer parameters in which to store that knowledge, so the natural prediction is a *larger* retrieval delta everywhere, and a larger gap between PyAether and SKILL. This is the test of whether "weights can replace retrieval"—the assumption a local subagent must rely on (Table XI).

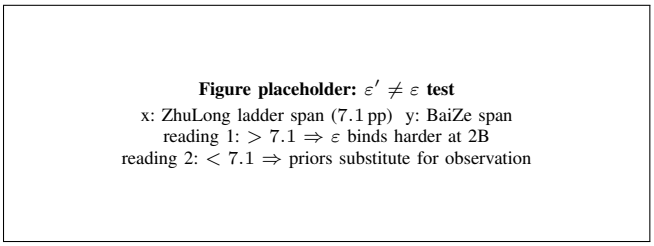


Fig. 3. The ε test of RQ4. Only the span is measured; both readings are fixed before the number exists, which is what keeps the test from becoming a post-hoc narrative.

D. RQ3: Is the Belief-Update Budget Still Monotone?

ZhuLong bounds the number of `run_code` calls per trace and finds accuracy monotone in k on PyAether (63.8% at k=1, 69.1% at k=3, 77.7% at k=10, unsaturated), with a *lagged* channel—returning the previous execution's state—landing between k=3 and k=10 at 73.0%. On a smaller model the same budget may bind earlier or later; both directions are informative, so we pre-register the shape, not the numbers (Table XII).

E. RQ4: Re-measuring the Read-Back Ladder on 2.2B (Claim C)

This is the load-bearing experiment of the paper. ZhuLong operationalizes observation fidelity as three closed-loop variants that differ only in what `run_code` returns: (N′) nothing, (B) pass/fail, (F) full state; pooled accuracy moves $69.8 \rightarrow 72.2 \rightarrow 76.9$, a span of 7.1 pp. Repeating that ladder on BaiZe produces one of two readings, and both are decisive:

- **Span** > 7.1 **pp** (expected): a model with weaker weight priors depends *more* on observation, so ε binds harder as scale falls; ZhuLong's ε-bound is then *strengthened* by the continuation rather than merely repeated.
- **Span** < 7.1 **pp**: the small model substitutes priors for observation, and the ε-bound is conditional on backbone capacity; the paper then reports the substitution ratio and revises the framing.

Table XIII therefore contains no hypothesis about *which* outcome obtains, only about what each outcome means.

F. Reward Design and Its Ablation

An execution sandbox turns correctness into a verifiable signal, so we optimise with group-relative policy optimisation and no value network—cheaper in memory, which matters at 2.2B on one node. Table XIV separates four rewards; a binary signal is the baseline, partial credit addresses its sparsity, a tool-call penalty aligns the policy with the harness cost model, and an API-legality term penalises hallucinated symbols. Because the reward is computed from assertions a policy could target directly, the anti-cheat hook stays on in every run and a cheat rate is reported beside every accuracy figure.

TABLE X

RQ1: POOLED AND PER-SLICE PASS@1 (%) OVER FIVE RUNS, MEAN±S.D., UNDER ZHULONG'S BYTE-IDENTICAL HARNESS (T6). ZHULONG ROWS ARE MEASURED; THE BAIZE ROW IS THE PRE-REGISTERED HYPOTHESIS. COST COLUMNS ARE PER TASK OVER ALL 449 TASKS.

Backbone	PyAether (158)	SKILL (157)	Tcl (134)	Overall (449)	Δ vs 2B	Time (s)	`run_code`	Input (k)
DeepSeek-V4-Pro	**82.5**±2.0	**60.3**±6.7	**89.9**±2.0	**76.9**±2.3	[TBD]	464	10.6	710
GLM-5.3-Flash	86.8±1.6	57.2±6.9	80.3±3.8	74.5±2.4	[TBD]	[TBD]	[TBD]	[TBD]
GLM-5.2	83.3±3.1	54.4±4.5	82.7±1.6	73.0±2.1	[TBD]	[TBD]	[TBD]	[TBD]
Kimi-K2.6	77.0±1.6	55.5±2.7	75.8±1.9	69.1±0.5	[TBD]	[TBD]	[TBD]	[TBD]
DeepSeek-V4-Flash	60.1±5.5	47.0±1.8	70.4±3.1	58.6±2.3	[TBD]	[TBD]	[TBD]	[TBD]
Doubao-Seed-2.0-Pro	63.8±1.4	42.5±3.4	53.3±3.4	53.2±1.9	[TBD]	[TBD]	[TBD]	[TBD]
BaiZe-2.2B	[TBD]	[TBD]	[TBD]	[TBD]	—	[TBD]	[TBD]	[TBD]

ZhuLong rows reproduced from its Table 4 (strongest–weakest span $76.9 - 53.2 = 23.7$ pp); cost columns are only reported for its primary backbone, so "[TBD]" there means *not measured by ZhuLong*, not *failed*. **Reverse contingency: if BaiZe <53.2% pooled, Claim A becomes a cost–accuracy Pareto statement and the paper reports the k=10 budget at which the gap closes (RQ3).**

TABLE XI

RQ2: FIVE HARNESS CONFIGURATIONS, POOLED PASS@1 (%) (T7). ZHULONG COLUMNS ARE MEASURED; THE BAIZE COLUMN IS THE HYPOTHESIS.

Configuration	PyA. (158)	SKILL (157)	Tcl (134)	ZL all (449)	BaiZe (449)	Δ
Full	82.5	60.3	89.9	76.9	[TBD]	[TBD]
w/o Retrieval	73.4	45.2	82.2	66.2	[TBD]	[TBD]
w/o Self-expl.	77.0	61.5	86.7	74.5	[TBD]	[TBD]
RAG (no sandbox)	71.1	52.2	84.6	68.6	[TBD]	[TBD]
Pure LLM	14.3	33.2	10.1	19.7	[TBD]	[TBD]

ZhuLong numbers from its Table 3. **Reverse contingency: if the BaiZe retrieval delta is *not* larger than ZhuLong's on SKILL, the "weights replace retrieval" hypothesis survives and the local-subagent argument strengthens; if it is larger, the deployment needs a retrieval stack after all, which is reported as a scope limit rather than hidden.**

TABLE XII

RQ3: BELIEF-UPDATE BUDGET ON PYAETHER, PASS@1 (%) (T8).

Budget	ZhuLong	BaiZe	Expected
k=1	63.8	[TBD]	↑ vs. k=0
k=3	69.1	[TBD]	monotone
k=10	77.7	[TBD]	best of the three
lagged (k=10)	73.0	[TBD]	between k=3, k=10
Converged (%)	99.1→76.3	[TBD]	tightens with k

Reverse contingency: if BaiZe saturates at k=3, the deployment defaults to three-round traces and the serving budget of Section VII is computed there; if it is still rising at k=10, the budget axis is the binding constraint and is reported as such.

G. Stage (v): Preference-Based Domain Adaptation

Stage (v) differs from stage (ii) in two respects, stated explicitly because otherwise the two coincide: the reward is learned from human judgement rather than verified, and the target behaviour is a team's internal conventions—naming rules, layer policies, review checklists—which no public benchmark measures. We collect pairwise judgements from in-house reviews over the stage-(iv) task families, fit a reward model, and optimise under a KL-regularised objective from the stage-(iv) checkpoint, since preference data is far scarcer than the verifiable signal. The rationale for a 2.2B core is strongest here and it is a constraint, not a preference: proprietary layouts and preference data cannot leave the premises, so the choice is a small local model or no model at all. Evaluation uses held-out review items scored by expert agreement, reported beside the execution-based numbers so that verified and preference-based adaptation are compared on one system.

VI. MULTIMODAL GROUNDING

The second goal is observational: layout and schematic views carry 70.2% of the benchmark's tasks, and waveforms—one of ZhuLong's four slices—are equally visual, yet its agent can only reach them through scripts. This section first fixes the encoder (stage iii, complete) and the alignment protocol (stage iv), then pre-registers three experiments on whether pixels, at a given token budget, actually widen or narrow the grounding gap.

A. Stage (iii): The Visual Encoder

We adopt **OpenVision2**, a pure-attention ViT at width-512/depth-30 (126.8M parameters), trained from scratch with a bidirectional trunk at 224px/16-px patches on a ≈18.5M-pair image–text corpus (CC12M + Amshaker). The objective combines a global InfoNCE term with a caption-*independent* masked-patch reconstruction term at 1:1 weight and a 0.6 mask ratio; a four-arm ablation at 15.4M pairs shows that neither term alone suffices (6.1% and 6.2% linear probe) while the combination reaches +6.0 pp, widening to +12.4 pp at 55.3M samples. On the frozen trunk, an ImageNet-1k linear probe gives 11.50% → **19.76**% top-1 from 5.1M to 55.3M samples and 20.27% at 176M samples (58.8M-pair corpus adding GPIC), with no saturation. The 224/16 setting yields 14×14=196 visual tokens, i.e. ∼16×16 px/token, and is retained because it is best on both loss and throughput. That choice was made for *general* images; whether it survives detail-dense line drawings is the question this section is built to answer, not to assume.

B. Stage (iv): Alignment Protocol

Alignment follows the open-MLLM curriculum—projector pre-training, mid-training, instruction tuning—with a lightweight projector connecting the encoder to the frozen-then-unfrozen backbone. Three task families are used, each chosen because it admits an objective ground truth: *image→script* (recreate the rendered view; supervision is the execution itself), *script→image* (predict the geometry a script produces, self-labelled the same way), and *layout VQA* (layer identity, device type, spacing, rule violations; expert-labelled). The first family is the strongest lever: its output is a script, so it is scored by the same assertion sandbox as stage (ii), which

TABLE XIII
RQ4: THE READ-BACK LADDER, N′ →B→F, PASS@1 (%) (T9). LEFT BLOCK MEASURED BY ZHULONG; RIGHT BLOCK IS THE PRE-REGISTERED RE-MEASUREMENT ON BAIZE.

Rung	PyA.	SKILL	Tcl	Pooled	BaiZe PyA.	BaiZe SKILL	BaiZe Tcl	BaiZe Pooled
(N′) no read-back	73.3	50.7	87.9	69.8	[TBD]	[TBD]	[TBD]	[TBD]
(B) binary pass/fail	78.1	57.2	83.0	72.2	[TBD]	[TBD]	[TBD]	[TBD]
(F) full observation	**82.5**	**60.3**	**89.9**	**76.9**	[TBD]	[TBD]	[TBD]	[TBD]
Span (F−N′)	9.2	9.6	2.0	**7.1**	[TBD]	[TBD]	[TBD]	[TBD]

Left block from ZhuLong Table 3 (lower block); the Tcl inversion at rung B is retained verbatim because it is the counter-example that makes the ladder a measurement rather than an assumption. **Reverse contingency: both directions are pre-registered (span larger ⇒ ε binds harder at 2B; span smaller ⇒ priors substitute for observation). No outcome leaves the claim untested.**

TABLE XIV
REWARD ABLATION IN STAGE (II) (T10).

ID	Reward	Pass@1	Cheat	Expected
R1	binary pass/fail	[TBD]	[TBD]	baseline
R2	R1 + assertion ratio	[TBD]	[TBD]	↑ vs. R1
R3	R2 + tool-call penalty	[TBD]	[TBD]	≈ acc., ↓ cost
R4	R3 + API legality	[TBD]	[TBD]	↓ hallucination

Reverse contingency: if R3/R4 lose accuracy, the cost penalty is over-weighted and is retuned rather than dropped, since the deployment objective is cost-aware by construction.

TABLE XV
RQ5: IMAGE→SCRIPT ASSERTION PASS RATE (%) BY VISUAL-TOKEN TIER AND VIEW TYPE (T11).

Tier	layout	schem.	wave	n	Expected
text-only	[TBD]	[TBD]	[TBD]	[TBD]	floor
T196	[TBD]	[TBD]	[TBD]	[TBD]	≤ floor
T576	[TBD]	[TBD]	[TBD]	[TBD]	↑ vs. T196
T1024	[TBD]	[TBD]	[TBD]	[TBD]	↑ or ≈ T576

Reverse contingency: if T196 is *not* worse than T576/T1024 (i.e. no threshold), Claim B's threshold clause is falsified and the paper reports that coarse representation suffices for EDA views—a null result with the same protocol.

TABLE XVI
RQ6: FINE-GRAINED DISCRIMINATION SUBSET (T12). TASK IDS ARE *to be instantiated* FROM ZHULONG'S TRACES.

ID	Task (failure class)	196	576	1024
F1	count nets in an Innovus view (composition)	[TBD]	[TBD]	[TBD]
F2	layer id of a drawn rectangle (API misuse)	[TBD]	[TBD]	[TBD]
F3	rising-edge count in a waveform (semantics)	[TBD]	[TBD]	[TBD]
F4	highlight vs. bounding box (API misuse)	[TBD]	[TBD]	[TBD]
F5	spacing violation present? (guard)	[TBD]	[TBD]	[TBD]
F6	device type from symbol (semantics)	[TBD]	[TBD]	[TBD]

Classes follow ZhuLong §6.3 verbatim so counts can be pooled: "composition"=API-composition error, "guard"=unauthorized guard, "semantics"=underspecified semantics, "API misuse"=wrong internal API.

TABLE XVII
RQ7: TEXT VS. VISION VS. BOTH, ASSERTION PASS RATE (%) (T13).

Channel	PyAether	SKILL	waveform	Expected
text only	[TBD]	[TBD]	[TBD]	baseline
vision only	[TBD]	[TBD]	[TBD]	↓ where docs matter
text + vision	[TBD]	[TBD]	[TBD]	↑ if ε shrinks

Reverse contingency: if text+vision does not beat text, the visual channel is not a fidelity gain at this encoder scale, and Claim B's positive half is reported as unsupported—while RQ6 still localizes where it fails.

lets multimodal and text-only configurations be compared on *one* metric instead of two incommensurable ones.

C. RQ5: How Many Visual Tokens Does a Line Drawing Need? (Claim B)

Claim B asserts a threshold rather than a suffices/not-suffices verdict: on detail-dense views, accuracy should be *increasing* in the token budget, with the 196-token tier at or below the text-only baseline. Table XV is the grid; each cell is an assertion pass rate from the image→script family, so the number is generated by the same sandbox that scores stage (ii).

D. RQ6: Fine-Grained Discrimination

The failure modes ZhuLong reports are exactly the ones a coarse visual representation should break on: a single rising edge returning `N/A`, net counts differing by three (1869 vs. 1872), a mandated call replaced by a visually plausible neighbour (`aeAddHighLightRect` vs. `GuCartBox(emyBoxF(...))`). We therefore build a discrimination subset (Table XVI) whose answers are small integers or symbols—quantities a 16×16 px/token grid must physically be able to resolve.

E. RQ7: Text, Vision, or Both?

The decisive question for the deployment claim is not whether vision helps in isolation but whether the *combination* beats text alone at equal task budget. Three configurations run on the same task set: script-only (text), image-only (pixels to a description to a script), and both (pixels plus docs, Table XVII).

F. Waveform Reading

Waveforms deserve their own paragraph because they are simultaneously semantically and geometrically dense: period, duty cycle, edge count, glitch width and threshold crossing are *measurements*, not labels. The encoder is a general-purpose $224/16$ ViT, so a 16×16 px/token grid must separate a real rising edge from a glitch that occupies a fraction of a pixel row—the same regime in which ZhuLong's WaveView slice already fails on a single edge (WaveView-037 returns `N/A`). Table XVIII fixes four subtasks and the three tiers; *no* claim is made in advance about whether 196 suffices.

VII. SERVING ECONOMICS

Claim A is an economic claim, and it cannot rest on architecture alone: a hybrid that decodes faster but costs more

TABLE XVIII
WAVEFORM READING SUBTASKS (T14), PASS RATE (%).

Subtask	T196	T576	T1024
period measurement (±5%)	[TBD]	[TBD]	[TBD]
edge count (exact)	[TBD]	[TBD]	[TBD]
glitch present (yes/no)	[TBD]	[TBD]	[TBD]
threshold-crossing time	[TBD]	[TBD]	[TBD]

Task source: ZhuLong WaveView slice (77 tasks) and the WaveView-037 single-edge failure; numeric tolerance follows the assertion sandbox, not a human scale.

TABLE XIX
SERVING ECONOMICS PER TASK (T15). LOCAL ROWS USE EQ. 2; THE API ROW IS PRICED AT THE PROVIDER'S RATE.

Configuration	Prefill (tok/s)	Decode (tok/s)	Mem (GB)	GPU-s /task	Rel. cost
BaiZe-2.2B (hybrid, local)	[TBD]	[TBD]	[TBD]	[TBD]	1.0×
dense-2.5B (local)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
API (ZhuLong, V4-Pro)	—	—	—	[TBD]	[TBD]

Throughputs must be measured on the production stack; the Megatron-Core direct-drive numbers used during pre-training (prefill 23,692 vs. 21,436 tok/s; decode 22.77 vs. 2.21 tok/s) are *not* used here because the dense decode figure is CPU-launch-bound and would overstate the gap. **Reverse contingency: if memory occupancy is negligible at the measured session lengths, Claim A must be stated on prefill/decode cost alone, and the architecture argument is reported as unrealised at benchmark scale.**

per task has not solved the deployment problem. This section fixes the accounting before any number is measured, so that the measurement cannot be chosen after the fact.

a) Accounting.: For one benchmark task with R `run_code` rounds, mean context L and G_{out} generated tokens, the local cost is

$$C_{\text{task}} = \underbrace{\frac{P_{\text{pre}}\, L\, R}{S_{\text{pre}}}}_{\text{prefill GPU-s}} + \underbrace{\frac{G_{\text{out}}}{S_{\text{dec}}}}_{\text{decode GPU-s}} + \underbrace{\frac{\mu\, M_{\max}\, T_{\text{wall}}}{M_{\text{gpu}}}}_{\text{memory occupancy}}, \qquad (2)$$

where $S_{\text{pre}}, S_{\text{dec}}$ are measured throughputs, $M_{\max}$ the peak session state of Eq. 1, and μ the amortised accelerator price; the API comparison uses the provider's per-token price at the same token counts. Memory occupancy is kept as a separate term precisely because it is the term a hybrid improves and a throughput-only comparison hides.

Two caveats are part of the claim, not footnotes. First, an API deployment carries no memory term but carries the provider's margin, so the comparison must price both at equal token counts rather than at list price. Second, the local model's advantage is bounded by utilisation: a single session on one GPU looks expensive until the same weights serve a team. We therefore report both $N{=}1$ and an $N{=}8$ multiplexed case, and mark the crossover as measured or unmeasured rather than asserted.

VIII. Error Analysis

Comparisons are only interpretable if failures are classified the same way, so we adopt ZhuLong's four classes verbatim—**wrong internal API** (a correct API family, but an undocumented binding contract violated), **unauthorized guard** (a call the runtime forbids in that context), **underspecified semantics** (the same name means different things across hosts), and **host-language contract** (the embedding language, not the EDA API, is violated). These are the L4 failures ZhuLong identifies as invisible in documentation and only weakly exposed by execution—precisely the regime where a 2.2B model with fewer API facts in weights should fail more, and where a visual channel either helps (if the violation is visible) or does not.

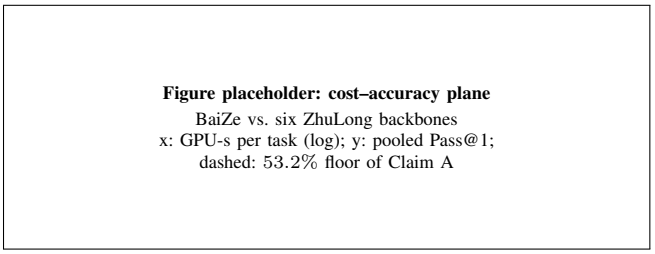


Fig. 4. Cost–accuracy plane on which Claim A is decided. ZhuLong backbone points are placed from Table X cost columns where available; the BaiZe point is the measurement of Table XIX.

One asymmetry is worth stating in advance: ZhuLong's SKILL slice has by far the highest failure denominator (31/281), because SKILL lacks the `apropos/describe` introspection that lets PyAether traces recover API knowledge from execution. If BaiZe fails disproportionately there, the honest conclusion is that a local subagent needs a retrieval stack on SKILL even if it does not on PyAether—a scope limitation we pre-commit to reporting.

IX. Discussion and Limitations

a) What would make this paper wrong.: Each claim has a stated failure condition, and we would rather name them than defend them: A fails if BaiZe pools below 53.2% or costs more per task than the API baseline; B fails if 196 tokens are not worse than 576/1024 on detail-dense views; C fails as a *test* if the ladder is not re-measured, not if its span comes out small—a small span is a result about substitution.

b) Limitations.: First, the visual study covers PyAether views; SKILL and Tcl visuals are not measured, so any claim that vision helps "EDA generally" is out of scope here. Second, 196 tokens is the operating point of a *general* encoder, chosen on loss and throughput rather than on EDA structure; a purpose-built encoder could move the threshold, and RQ5 measures the threshold rather than optimising it. Third, the interaction between the RL reward and the anti-cheat hook is unverified: a policy optimised against assertions may learn to satisfy the hook rather than the task, and while we report a cheat rate beside every accuracy, we have not shown that the rate is stable under RL pressure. Fourth, the serving numbers are single-node; multi-tenant amortisation is modelled but not measured.

c) If the 2B model collapses on SKILL.: The subagent proposition would then hold only for PyAether—a toolchain with introspectable APIs, where execution can substitute for retrieval. That is a narrower but still publishable claim, and

TABLE XX
ERROR TAXONOMY POOLED WITH ZHULONG (T16). COUNTS ARE EXECUTION-STAGE FAILURES OVER ALL RUNS; ZHULONG TOTALS FROM ITS §6.3.

Class	ZL PyA.	ZL SKILL	ZL WaveView	ZL Innovus	BaiZe PyA.	BaiZe SKILL	Δ
Wrong internal API	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Unauthorized guard	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Underspecified semantics	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Host-language contract	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Total execution failures	29/109	31/281	1/43	1/23	[TBD]	[TBD]	[TBD]

ZhuLong reports totals only (29 of 109 PyAether traces, 31 of 281 SKILL, 1 of 43 WaveView, 1 of 23 Innovus); the per-class split of those totals is re-derived from the same traces so the two papers can be pooled cell by cell. **Reverse contingency: if BaiZe's failures concentrate in "wrong internal API" rather than in L4 classes, the diagnosis is missing knowledge rather than insufficient observation, and the remedy is retrieval/annealing, not pixels.**

it would sharpen rather than weaken ZhuLong's toolchain-specific finding: component contributions differ by toolchain, so the *deployable* subset of the harness differs by toolchain too. We pre-commit to reporting this boundary instead of averaging it away in the pooled number.

X. CONCLUSION

BaiZe is a from-scratch system—2.22B hybrid language backbone, 126.8M visual encoder, five stages—built to answer two questions that ZhuLong's observation-fidelity account leaves open: whether a local 2.2B model can reach the accuracy floor of the weakest measured large backbone at lower serving cost, and whether a visual channel shrinks or widens the grounding gap on detail-dense EDA views.

We report status rather than a verdict. Stages (i) and (iii) are complete and measured: 2.94T pre-training tokens at iteration 30,660 with loss 2.151 and ≈598K tok/s, and an encoder whose frozen-trunk ImageNet probe goes from 11.50% to 19.76% and 20.27% at 55.3M and 176M samples. Claims A, B and C remain **unverified**; their placeholder tables (Tables X–XVIII, XIX–XX) carry the hypotheses, the control configurations and the reverse contingencies under which each would be reported as false.

The single experiment that makes this a continuation rather than an application is RQ4: the read-back ladder measured on the same benchmark, with a span compared against 7.1 pp. A larger span would say that as models shrink, observation binds harder—ZhuLong's conclusion, extended rather than repeated; a smaller span would say that weight priors can partly pay for observation, which is a different theory of the same data and would be reported as such. We release the pre-training recipe, the encoder recipe, the message schema and the harness checklist so that either outcome is reproducible by a third party.

APPENDIX

Table XXI is the execution order; it exists so that a reader can tell which claims are cheap to check and which are blocked. Table XXII maps every claim to its research question, its table, and its falsifier, so that no placeholder table can be read as a result.

a) Six-page version.: If the venue cap is six pages, the cut is mechanical: keep Tables X, XIII, XX; move II–IV, XI–XIV, XV–XVIII, XIX, XXI and XXII to a supplementary file; keep §I, §V-B, §V-E, §VI-C–§VI-F and §IX in the body.

TABLE XXI
EXPERIMENT PLAN BY PRIORITY (P0 FIRST). BUDGETS ARE ESTIMATES ON 8×H100 UNLESS STATED.

Exp.	Depends on	Shape	Est.	P
RQ1	stage (ii) SFT	449×5 runs	≈290 GPU-h	P0
RQ4	RQ1 harness	3×158×5	≈60 GPU-h	P0
RQ2	RQ1 harness	5×449×5	≈1,450 GPU-h	P1
RQ5/RQ6	stage (iv)	50×3 tiers	≈40 GPU-h	P1
T15	hardware	measurement	≈2 GPU-h	P1
SKILL/Tcl	RQ1 pass	2×314×5	≈800 GPU-h	P2
RQ7	RQ5/RQ6	3× channels	≈120 GPU-h	P2
RQ3/T10	RQ1 harness	4×158×5	≈200 GPU-h	P3

Estimates scale ZhuLong's measured 464 s/task and five-run protocol; they exclude data preparation and are marked *est.*. A blocked row is reported as blocked, never as zero.

The four-stage pre-training and encoder sections compress to their tables, since both are complete and already reported elsewhere.

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TABLE XXII

CLAIM → RQ → TABLE → FALSIFIER INDEX. EVERY TABLE IN THE PAPER APPEARS EXACTLY ONCE.

Claim	RQ	Table	Question	Falsifier / reverse contingency
A	RQ1	X	pooled $\geq 53.2\%$?	$< 53.2\% \Rightarrow$ Pareto report, not a claim
—	RQ2	XI	does 2B need retrieval more?	delta not larger $\Rightarrow$ weights substitute
—	RQ3	XII	is k-monotonicity kept?	saturates at $k{=}3 \Rightarrow$ default budget 3
C	RQ4	XIII	ladder span vs. 7.1 pp	both spans pre-interpreted
—	RL	XIV	which reward wins?	cost penalty over-weighted $\Rightarrow$ retune
B	RQ5	XV	is there a token threshold?	no threshold $\Rightarrow$ coarse suffices
—	RQ6	XVI	what breaks at 196?	failures localize the mechanism
—	RQ7	XVII	does text+vision beat text?	no $\Rightarrow$ visual gain unsupported
—	wave.	XVIII	does 196 read edges?	source: WaveView-037
A	econ	XIX	cost per task	memory negligible $\Rightarrow$ restate A
—	err.	XX	four-class split	failures not L4 $\Rightarrow$ remedy is retrieval
—	arch	II, III, IV	protocol / tiers / identity	any deviation invalidates comparison

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